Network (Tree) Topology Inference Based on Prüfer Sequence

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***Abstract*— Network topology discovery is the basis for any network management application. The problem of estimating internal structure and link-level performance from end-to-end measurements is known as network tomography. This paper proposes a novel approach to discover network characteristics, in particular, tree topology from the hop count metric (distance) between OD (Origin – Destination) pairs. The proposed method is based on Prüfer encoding and decoding techniques of trees using this metric. The method also has the potential to minimize and avoid reliance on ICMP.**

Keywords-Measurement; Network Tomography, Path, Prüfer sequence, Topology inference, Traffic flow, ICMP

I. INTRODUCTION

At the advent of advanced networking services such as Telepresence, IPTV and mobile interactive applications, it is critical that all components in the overall services delivery system provide comprehensive end-to-end quality of experience for all subscribers. This application layer requirement can be eventually mapped to a network layer requirement. The problem of estimating internal network structure and link-level performance from end-to-end measurements is called network tomography. This paper describes network topology discovery from the path metrics between Origin-Destination (OD) pairs of a network. Inferring network topology by network discovery is the critical step in network tomography [2] and is essential to any network management applications. This paper focuses on the first step of determining the links in the network, given end to end measurements, prior to deriving link level metrics by Network Tomography.

One of the most extensively used methods for IP network topology discovery is skitter [5], to obtain the router-level topology of the Internet. Traceroute was used as the probe. References [6] and [7] discuss improvements to the traditional "traceroute" to reduce redundant probing. Donnet et al [6] propose to improve on skitter by introducing the DoubleTree algorithm to reduce redundancies in probing from monitors to destinations. The key ideas were to exploit the tree-like structure of the routes to and from a single point to help decide when probing should stop. Further efficiencies are discussed in [7], introducing the BackTrace Tree algorithm, which exploits the technique of using the estimate of distance to destination to backtrack from the furthest routers, resulting in large efficiency gains.

Note that the method in [7] needs to record the interfaces of the discovered routers and stop the probing against the destination when it finds that a router interface has already been probed.

This paper assumes the following:

1. We are interested in discovering the logical topology instead of physical topology. Physical topology represents the topology model of layer-1 (the physical layer) whereas the logical topology represents the topology model at layer-2 or layer-3 of the OSI model. The logical topology discovery can be done in two different ways: (i) setting up probes in the network, collecting management information at internal nodes periodically and generate topology and report other network management related information. (ii) Inferring the topology and other network characteristics from end-to-end measurements. We will focus on approach (ii) to infer the topology.
2. The underlying network is actually a *tree* at any instance of communication. It means that there exists a unique path between any OD pairs in the network at a given point of time.
3. The existing path discovery tools such as *traceroute*, *ping* will provide the path metrics between any OD pair in the network.
4. The network considered is *undirected*. It means that the forward and reverse flows are through the same link, though in reality it will be through different links. The proposed approach can be extended to directed network.
5. The network considered is connected or strongly connected if there exists a path between any pair of nodes in the network.

The paper is organized as follows. In section II the problem is defined. Path metric between OD pairs is defined in section III. Section IV describes Prüfer encoding and decoding. Section V proposes topology discovery algorithm using Prüfer encoding technique from path metric. An example and proof for the theorem is shown in section VI. Section VII concludes the paper.

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II. PROBLEM STATEMENT

Graphs are the mathematical constructs best suited to represent networks. A network in its simplest form is a set of nodes or vertices joined together in pairs by edges. In the context of communication networks, vertices are Autonomous Systems, routers, switches, computers, work stations whereas the edges are links connecting a pair of such vertices. A set of connected links leading from an origin node to a destination node is referred to as a "path". Consider a fixed or static routing protocol in that for each OD pair, the same unique path always carries the traffic between origin and destination and vice-versa. The path length or OD distance is measured as the number of links connecting an OD pair.

The problem is to infer the topology (tree) and its internal nodes given the path measurements or distance between every OD pairs in a network.

III. PATH METRIC

Let $T = (V, E)$ denote the tree with node set V and link set E. Let $R \subset V$ denote the set of leaf nodes in the tree. Let R x R represent distinct OD pairs of the network. A walk in a graph is a finite non-null sequence $W = v_0e_1v_1e_1v_2...e_kv_k$, whose terms are alternately vertices and edges, such that, for $1 <= i <= k$, the ends of e_i are v_{i-1} and v_i. In other words, a walk of length k is a sequence of vertices $v_0, v_1, ... v_k$ such that v_i is adjacent to v_{i+1} for each i. If the edges $e_1, e_2, . . . , e_k$ of a walk are distinct, it is called a *trail*. If the vertices $v_0, v_1, . . . , v_k$ are distinct, it is called a *path*. A *closed walk* is a walk of length k such that $v_0 = v_k$. A *cycle* is a closed walk where none of the vertices repeat except for the first and the last. The *degree* of a vertex is defined as the number of vertices its adjacent to, or the number of edges incident with it. Refer [1].

A graph is said to be *connected* if for all pairs of vertices there exists a walk. A *tree* on n vertices is a connected graph that contains no cycles. The distance matrix or path length matrix D for OD pairs is defined as follows:

Entries $d_{ij} = d_{ji}$ = path length between OD pair i and j, where i and j ranges from 1 to $|R|$. $d_{ii} = 0$. Refer [3]. This matrix is a symmetric matrix.

This path metric could be obtained using tools like *traceroute*. The "traceroute" program verifies whether an end-to-end IP path is operational, and provides information on the intermediate systems to be found along the IP (Layer-3) path from the origin to the destination. Traceroute uses ICMP echo response with different TTL (Time-To-Live) packets to determine the path. In reality, however, some routers discard ICMP (Internet Control Message Protocol) messages and thus make discovery incomplete. In such cases, our approach of determining the topology using only the hop counts between each pair is very helpful, since this requires only a single UDP request/response sequence, with the TTL in the arrived response packet providing the hop count. Although traceroute was used to determine the hop count in our work, it can easily be replaced by such an UDP probe. A further improvement would be to use a TCP probe and rely on the RST or SYN/ACK received from the destination, rather than on the ICMP Destination unreachable message.

IV. PRÜFER SEQUENCE

A Prüfer sequence is a sequence of n-2 numbers or labels, each being one of the numbers 1 through n, with repetitions allowed. It should be noted that there are n^{n-2} Prüfer sequences for any given n. This can be easily proved: by definition, there are n ways to choose each element of Prüfer sequence of length n-2. Since there are n-2 elements to be determined, in total we have n^{n-2} ways to choose the whole sequence. Given a labeled tree with vertices labeled by $1,2,3,...,n$, the Prüfer encoding algorithm outputs a unique Prüfer sequence of length n-2. The Prüfer decoding algorithm provides the inverse algorithm, which finds unique labeled tree T with n vertices for a given Prüfer sequence of n-2 elements. Refer [4].

Key observation:

1) Any vertex v of T occurs d(v)-1 times in Prüfer sequence P, where d(v) is the degree of vertex v.
2) The vertices of degree 1, i.e., the leaves, in T never appear in P.

V. TOPOLOGY DISCOVERY

In our problem the key assumption is that we have path length between all OD pairs in the network represented as a matrix. The OD pairs denote the leaves in the underlying tree T. The key fact is that there exists a unique tree T for the given path matrix between all its leaves. This can be easily shown that if there is another path length between a given OD pair, then there is a cycle, which is a contradiction by the definition of tree. The idea is to relate this uniqueness relationship between path matrix, Prüfer sequence and labeled tree.

The leaf nodes are labeled by 1, 2, 3, …, r and denoted by the set R. The problem is to find the exact number of internal nodes connecting these leaf nodes and derive the Prüfer sequence.

Since the lowest label among the leaves is well defined, we consider m to be the first internal node connecting to the lowest label. The label m is r+1. Thus we generate unique labels in the increasing order for every internal node discovered. Let us consider the lowest label as l. Now m is in the path of all OD pairs that has l as origin or destination, and subsequently the path length to and from m is reduced by one as compared to path lengths to and from l. As per Prüfer encoding, l can removed and its unique neighbor m is added to the Prüfer sequence. The vertex m is also added to the path matrix with appropriate path lengths to the rest of the vertices.

Now there are two possible scenarios with respect to m:

1) m is connected to another leaf node with lowest label (other than l). This can be seen by checking if there exists a path with length 1 between a leaf node and m.
2) m becomes a leaf node after removal of l.

This procedure is repeated with the next lowest label in R.

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Once all leaves are considered, the set of newly discovered nodes, say S form two partitions viz. leaves and intermediate nodes. The new set of leaves R' from S is determined by checking the transitive relationship between all newly discovered nodes. Since removal of a leaf node in a tree will not change the properties of the tree, the next lowest labeled node is well defined from this new set R'. This procedure is repeated with R'. The algorithm stops when there are exactly two nodes with path length 1.

A. *Prüfer Toplogy Discovery*

Algorithm: Prüfer Topology Discovery

Input: Leaf nodes $R = (1, 2, 3, \ldots, r)$

Path Matrix $D[1..r, 1..r]$

Output: Prüfer sequence P

```
last_node = r; P = {};
repeat:
  while (R not empty)
  begin
    S = {}
    k = leaf with the lowest label from R
    if k has a neighbor in P, say P_i,
    then
      begin
        m = P_i
        Append m to Prüfer sequence P
        Remove k from path matrix
        if there is only one d_ij == 1 in D, goto done:
      end
    else
      begin
        increment last_node
        m = last_node
        Consider m as the neighbor of k
        Append m to Prüfer sequence P
        Remove k from path matrix
        Add m to the path matrix
          d_im = d_mi = d_ik – 1,  i belongs to {1, 2, .. m-1}
        Append m to S
      end
  end
  R' = getleaves(S, D)
  if |R'| == 2 and d_ij == 1, goto done:
  else R = R'; goto repeat:
done: Print P
```

getleaves is a procedure that partitions the intermediate nodes generated into set of leaf nodes and non-leaf nodes after removing the previous set of leaf nodes. It is determined by checking the transitive relationships (triangular inequality) between the generated intermediate nodes. That is, if a path between a pair of vertices u and v passes thru an intermediate vertex w, then dist(u, v) = dist(u, w) + dist (w, v). A node m is considered as leaf node if m does not appear as an intermediate node between any pair of vertices discovered so far.

B. *Discovering leaves*

Algorithm: Get Leaves

Input: Intermediate nodes Set *S* and Path matrix *D*

Output: Set of leaf nodes

```
Mark all nodes in S as leaf nodes
for i ranges on |S|
    for j ranges on |S|
        for k ranges on |S|
            if d_ij == d_ik + d_k then mark node k as non-leaf
return all nodes that are marked as leaf nodes in S.
```

The correctness of the algorithm is based on the proof of correctness of Prüfer encoding and decoding algorithms and the fact of unique tree representation for the given path matrix and Prüfer sequence.

C. *Time Complexity*

The time complexity of this algorithm is derived as follows:

The algorithm begins with R leaf nodes. After processing the first set of R leaf nodes, R' will be at least 2 and at most R. This is due to the fact that after removing all leaf nodes, say R in a tree, the remaining tree will have at most R nodes and at least 2 nodes. The complexity for inserting an intermediate node will be $O(R)$, and the complexity for the procedure *getleaves* will be $O(R^3)$. At the end of each iteration, the path length is guaranteed to reduce by 2, as at least 2 leaves are removed. The procedure stops when there are only 2 leaves remaining and the distance between them is 1. So, considering the maximum path length as L in D, the algorithm takes $L/2*R^3$ steps to finish, and hence the complexity is $O(R^3)$.

VI. EXPERIMENTAL RESULTS

The following path matrix D was obtained for a simulated network where ICMP enabled to discover Layer-3 distance metric. The network topology, configuration and further details on the approach are available for reference at [8].

$D[1..8,1..8] =$

$$\begin{matrix} 0 & 2 & 2 & 4 & 5 & 6 & 6 & 4 \\ 2 & 0 & 2 & 4 & 5 & 6 & 6 & 4 \\ 2 & 2 & 0 & 4 & 5 & 6 & 6 & 4 \\ 4 & 4 & 4 & 0 & 3 & 4 & 4 & 2 \\ 5 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 6 & 6 & 6 & 4 & 3 & 0 & 2 & 4 \\ 6 & 6 & 6 & 4 & 3 & 2 & 0 & 4 \\ 4 & 4 & 4 & 2 & 3 & 4 & 4 & 0 \end{matrix}$$

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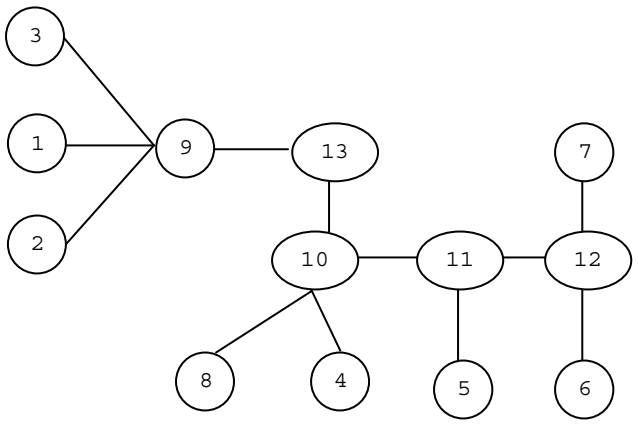

$P = (9\ \ 9\ \ 9\ \ 10\ \ 11\ \ 12\ \ 12\ \ 10\ \ 13\ \ 11\ \ 10)$

Figure 1

Theorem 1

The above algorithm produces a unique Prüfer sequence associated with the path matrix D[1 .. r, 1 .. r], where r is the number of leaf nodes for a tree T.

***Proof*:**

Let $T = (V, E)$ be a tree, where $V = \{1, 2, .. n\}$ and E is a set of n-1 edges.

Let A be the adjacency matrix that represents the tree T. The adjacency matrix of a tree $T = (V, E)$, where $V = \{1, 2,\ldots, n\}$, is the matrix $A[1 . . n, 1 . . n]$ given by

$A[i, j] =$ 1 if (i, j) belongs to E

0 if (i, j) does not belong to E.

Let $R = \{1, 2, 3, .. r\}$ be the set of initial leaf nodes in our algorithm. Let $K = \{m, m+1, m+2, .. m+k-1\}$ be the set of internal nodes, where $m = r+1$. $|R| = r$ and $|K| = k$, $r \geq 2$, $k \geq 1$. This implies the total number of nodes $n = r + k$.

Let P and Q be the two different Prüfer codes of length n-2 obtained from the path matrix D using our algorithm. P and Q are Prüfer codes of any sequence of labels taken from k labels in K, with repetitions allowed and any member of K appears at least once.

Since the codes P and Q differ in at least one position and also any OD pair there is a unique path, it suffices to show that any change in the code produces a different adjacency matrix, which in turn produces a different distance matrix.

Let *last*[1 .. n-2] be a linear array that marks the rightmost (*last*) occurrences of labels in the code. The value of *last*[i] is 1 if i is the rightmost position in the code for the node in $P[i]$ and it is 0 otherwise.

Let R' be the new set of leaf nodes, which has been obtained from the removal of previous set of leaf nodes. This set is created by adding $P[i]$ to R' when *last*[i] = 1, for the previous string of sequence scanned in the code.

Initially, the adjacency matrix $A[1..n][1..n]$ is initialized to zero. The entry $A[i, j]$ is set to 1 for the following conditions:

When i belongs to R,

The first r nodes in the Prüfer code are incident to leaf nodes in R.
$A[i, P[i]] = 1$, where $i = 1, 2, \ldots, r$

When i belongs to K,

Consider R', which is the new set of leaves, such that $2 \leq |R'| \leq |R|$

$A[i, j] = 1$, where i belongs to R' and $j = P[t]$, where $P[t]$ is adjacent to i and t indexes into corresponding labels in the Prüfer code starting from $|R| + 1$, but up to $|R| + |R'|$. Note that $|R| < t \leqslant |R| + |R'|$

R' is computed for each iteration, and R becomes $R + R'$. This is repeated until the Prüfer sequence P and Q are completely scanned. This will generate $n - 2$ edges, since there are $n - 2$ labels in the Prüfer sequence. There are two possible scenarios in determining the last edge:

1. R' has only one element, say u, left with no adjacency element left in P and Q. Let v be the last element in the Prüfer code. In this case, (u, v) forms the last edge: $A[u,v]$ is set to 1.
2. R' has two elements, say u and v, but no adjacency element left in P and Q. In this case, (u, v) in R' forms the last edge: $A[u,v]$ is set to 1.

There are 2 cases wherein the codes P and Q differ:

Case *I*

The codes P and Q differ in any one position in the first r positions in the code. In this case, the leaf nodes will have different adjacency nodes, and hence j will be different in $A[i, j]$. This will produce different adjacency matrices for P and Q. In other words, j will be same if and only if the code P and Q are same.

Case *II*

The codes P and Q differ in any one position in the remaining n-r-2 positions in the code. In this case, the set R' will be different for both codes in at least one iteration, which in turn will have different values for i and j to make $A[i,j] = 1$. Iteration is defined as discovering a set of leaf nodes from the *last* array, and associating neighbors from the code as mentioned above.

Hence in both cases, change in any one position of the codes will have a change in the adjacency matrix, and hence change in the structure of the tree.

Hence P and Q should be identical if the above procedure needs to generate the same adjacency matrix.

Hence the proof. ■

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The above algorithm is verified by computing the distance between all pairs of vertices using Floyd Warshall algorithm on the adjacency matrix obtained from the Prüfer code.

The adjacency matrix for the Prüfer code obtained in the above example (See figure 1):

P = {9 9 9 10 11 12 12 10 13 11 10}

A[1..13,1..13] =

$$\begin{matrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0
\end{matrix}$$

Applying Floyd Warshall algorithm on the above adjacency matrix produces the following distance matrix that represents the path length between every pair of vertices:

D[1..13,1..13] =

$$\begin{matrix}
0 & 2 & 2 & 4 & 5 & 6 & 6 & 4 & 1 & 3 & 4 & 5 & 2 \\
2 & 0 & 2 & 4 & 5 & 6 & 6 & 4 & 1 & 3 & 4 & 5 & 2 \\
2 & 2 & 0 & 4 & 5 & 6 & 6 & 4 & 1 & 3 & 4 & 5 & 2 \\
4 & 4 & 4 & 0 & 3 & 4 & 4 & 2 & 3 & 1 & 2 & 3 & 2 \\
5 & 5 & 5 & 3 & 0 & 3 & 3 & 3 & 4 & 2 & 1 & 2 & 3 \\
6 & 6 & 6 & 4 & 3 & 0 & 2 & 4 & 5 & 3 & 2 & 1 & 4 \\
6 & 6 & 6 & 4 & 3 & 2 & 0 & 4 & 5 & 3 & 2 & 1 & 4 \\
4 & 4 & 4 & 2 & 3 & 4 & 4 & 0 & 3 & 1 & 2 & 3 & 2 \\
1 & 1 & 1 & 3 & 4 & 5 & 5 & 3 & 0 & 2 & 3 & 4 & 1 \\
3 & 3 & 3 & 1 & 2 & 3 & 3 & 1 & 2 & 0 & 1 & 2 & 1 \\
4 & 4 & 4 & 2 & 1 & 2 & 2 & 2 & 3 & 1 & 0 & 1 & 2 \\
5 & 5 & 5 & 3 & 2 & 1 & 1 & 3 & 4 & 2 & 1 & 0 & 3 \\
2 & 2 & 2 & 2 & 3 & 4 & 4 & 2 & 1 & 1 & 2 & 3 & 0
\end{matrix}$$

From the above matrix, it is seen the sub matrix D[1..8,1..8] is the actual distance matrix between leaf nodes.

Our approach is to follow the idea of [7] in determining the path length to destination from the decremented TTL value in the response from the destination. The internal topology is then determined using the algorithm described above, rather than being built link by link as discussed in all of the literature cited. This should yield large benefits in terms of eliminating redundant probe IP packets, since a single UDP or TCP probe is sufficient for each OD pair to determine the entire internal topology of the network. Where nodes are known without their interface IPs, supplemental probes with appropriate TTL values can be used to determine the IP. A single such probe would be sufficient for each such unknown IP.

VII. CONCLUSION

In this paper we have proposed a new algorithm to discover the topology (tree) using Prüfer code from the distance matrix (distance between leaf nodes). We also verified the correctness of this algorithm by deriving the adjacency matrix from the Prüfer code and computing the distance between every pair of vertices using Floyd Warshall algorithm. Note that traceroute were used to determine the hop count, rather than a single UDP probe which would have been sufficient, as described earlier.

The focus of the paper was in discussing the algorithm that would enable computation of topology by determining only the path distance, rather than determining it link by link. We feel this approach is novel in this area, with potential for substantial efficiency in probes, especially in topologies typical of applications such as IPTV. In such scenarios, hardware probes are typically deployed at the edges of the network, making the approach feasible. Further work can implement the simple UDP or TCP probe and compare the efficiency with the method in [7]. Note that our approach may not rely on ICMP for the discovery of topology. It may be required for determining interface IPs, as one of the possible approaches.

It will be interesting to study the behavior of the algorithm when the underlying topology is a graph.

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